

- Ready-made plugins are available for public cloud vendors (like AWS, Azure, Google Cloud, Alibaba Cloud and Salesforce) to interact with the hypervisor.
- The key benefits of Nagios are:
 - Increased server and services availability
 - Fast detection of network outages
 - Fast detection of cloud computing environment problems

ManageIQ

ManageIQ is an open source platform for hybrid IT that helps to manage small and medium scale environments. It also has the capability to manage VMs, containers and public cloud resources. It has an agentless approach to collect the data. As it supports REST and SOAP interfaces, we can interact with any API supported application or tool. Refer to Figure 6 for a sample dashboard.

Its key benefits are:

- Continuously monitors the latest state of the resource
- Deploys a self-service UI for end users
- Maintains compliance across environments

- Optimises the usage of the environment
- The best part of ManageIQ is that it is distributed as a fully installed virtual machine and comes with preconfigured CentOS 7.5, PostgreSQL 9.5, Rails 5, the ManageIQ application and all associated Ruby gems pre-installed. The appliance is available for VMware, Red Hat Enterprise Virtualisation and OpenStack. Appliances support horizontal scaling as well as dividing up work by roles. An appliance can be configured to handle work for one or many roles, with workers within the appliance carrying out the duties for which they are configured.

Multiple appliances can logically group into zones. The zones are configured to provide specific functionalities like UI Zone, Production Zone, etc. ManageIQ has the option to create zone affinity for a provider. A region is a full installation of ManageIQ, containing one database appliance and potentially many other appliances. It is the collection of all zones that share the same database. In a typical enterprise installation, a separate region is used for each geographical region. For example, for an international corporation,

one region may be placed in North America, a second in Europe and a third in Asia. When multiple regions are involved, we refer to the collection of all regions as the enterprise, like in Figure 7.

The point to note here is that only one appliance per region can provide the database. The ManageIQ appliance comes preconfigured with a default database. If a second appliance is added, it must be configured to point to the first appliance's database.

Cloud Custodian

Cloud Custodian is another open source tool to monitor IT resources along with security and compliance. This tool is the right example of a script/rule based engine for monitoring cloud resources.

Cloud Custodian assists in good management in the cloud. The YAML DSL script allows you to define rules to enable a well-managed cloud infrastructure, which is both secure and cost optimised. It consolidates many of the *ad hoc* scripts that organisations have for lightweight and flexible tools, with unified metrics and reporting.

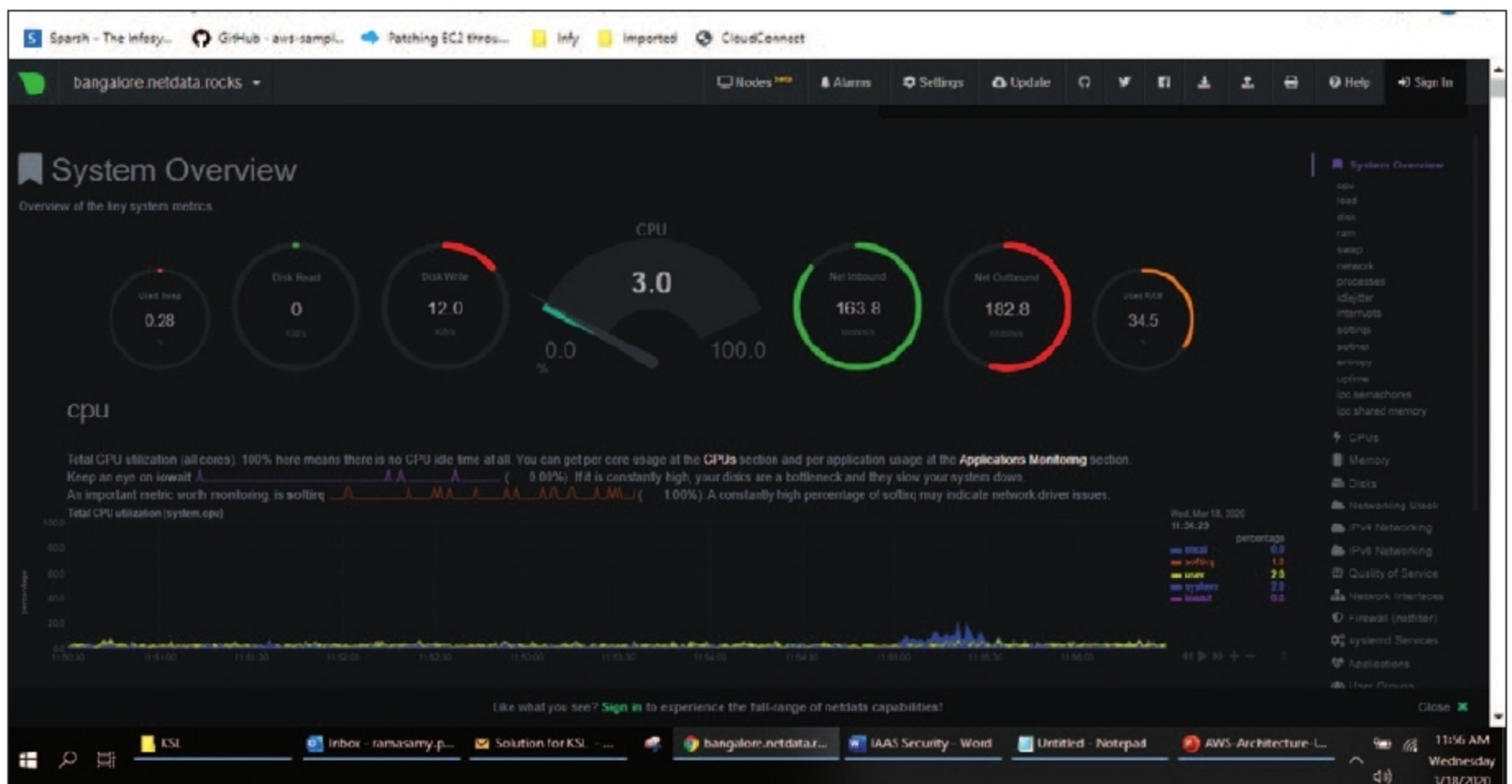


Figure 8: Netdata